King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 201 Major Exam II Term 252

Original exam date: 14 April 2026. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. The range of the function $f(x, y) = \ln\left(25 - x^2 - y^2\right)$ is

(a) $(-\infty,\, 2\ln 5]$

(b) $(-\infty,\, 5\ln 2]$

(c) $(-\infty,\, \infty)$

(d) $(0,\, 5\ln 5]$

(e) $[2\ln 5,\, \infty)$

2. If z is defined implicitly as a differentiable function of x and y by the equation

$$3x^2z - x^2y^2 + 2z^3 + 3yz - 5 = 0,$$

then $\dfrac{\partial z}{\partial x}$ is equal to

(a) $\dfrac{-2x(y^2 - 3z)}{3(x^2 + 2z^2 + y)}$

(b) $\dfrac{2x(y^2 - 3z)}{3(x^2 + 2z^2 + y)}$

(c) $\dfrac{2xy^2}{3(x^2 + 2z^2 + y)}$

(d) $\dfrac{2x^2y - 3z}{3(x^2 + 2z^2 + y)}$

(e) $\dfrac{2(xy^2 - 3xz - 3z^2)}{3(x^2 + y)}$

3. The graph of the equation $x^2 - 2y^2 + z^2 - 4x + 4y - 6z = k$ represents

(a) an elliptic cone if $k = -11$.

(b) an elliptic cone if $k = 20$.

(c) an ellipsoid if $k = 1$.

(d) a hyperboloid with two sheets if $k = 0$.

(e) a hyperboloid with one sheet if $k = -20$.

4. If θ is the angle of inclination of the tangent plane to the surface $3x^2 + 2y^2 - z = 4$ at the point $(1, 1, 1)$, then $\cos\theta =$

(a) $\dfrac{6}{\sqrt{53}}$

(b) $\dfrac{1}{\sqrt{11}}$

(c) $\dfrac{1}{2\sqrt{13}}$

(d) $\dfrac{1}{\sqrt{53}}$

(e) $\dfrac{4}{\sqrt{53}}$

5. If the symmetric equations of the normal line to the surface $z = ye^{2xy}$ at the point $(0, 2, 2)$ are given by

$$\frac{x}{8} = \frac{y-2}{b} = \frac{z-2}{c},$$

then $b + c =$

(a) -2

(b) 1

(c) -1

(d) 0

(e) 2

6.

$$\lim_{(x,y)\to(0,0)} \frac{x-y}{\sqrt{x}-\sqrt{y}} =$$

(a) 0

(b) 1

(c) 2

(d) does not exist

(e) $\sqrt{2}$

7. The directional derivative of $f(x, y) = x^2 \sin 2y$ at the point $\left(1, \frac{\pi}{2}\right)$ in the direction of the vector $\mathbf{v} = 3\mathbf{i} - 4\mathbf{j}$ is equal to

(a) 8

(b) -8

(c) $-\frac{8}{5}$

(d) $\frac{6}{5}$

(e) $\frac{8}{5}$

8. The total resistance R (in ohms) of two resistors connected in parallel is given by

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}.$$

By using the differentials dR, the change in R as R_1 is increased from 10 ohms to 10.5 ohms and R_2 is decreased from 15 ohms to 13 ohms, is approximately equal to

(a) -0.39

(b) 0.39

(c) -1.40

(d) 0.14

(e) -0.14

9. Let $f(x, y, z) = ye^x + x \ln z$. Then $f_{xzz}(1, 1, 1) + f_{zxz}(1, 1, 1) + f_{zzx}(1, 1, 1) =$

(a) 1

(b) -27

(c) -1

(d) -3

(e) -8

10. The distance between the point $(2, 8, 4)$ and the plane $2x + y + z = 5$ is equal to

(a) $\dfrac{11}{2}$

(b) $\dfrac{16}{\sqrt{6}}$

(c) $\dfrac{11}{2\sqrt{21}}$

(d) $\dfrac{11}{\sqrt{6}}$

(e) $\dfrac{21}{\sqrt{6}}$

11. If the equation of the tangent plane to the surface $z^2 - 2x^2 - 2y^2 = 12$ at the point $(1, -1, 4)$ is given by $x + ay + bz + c = 0$, then $a + b + c =$

(a) 4

(b) -2

(c) 2

(d) 3

(e) 5

12. If $w = xy + yz + xz$, where $x = s\cos t$, $y = s\sin t$ and $z = t$, then $\dfrac{\partial w}{\partial s}$ when $s = 1$ and $t = 2\pi$ is equal to

(a) 2π

(b) 2

(c) 0

(d) $1 + 2\pi$

(e) 4π

13. The maximum value of the directional derivative of $f(x, y) = y^2 - x\sqrt{y}$ at the point $(0, 3)$ is equal to

(a) $\sqrt{45}$

(b) 39

(c) 33

(d) $\sqrt{33}$

(e) $\sqrt{39}$

14. If the parametric equations of the line passing through the point $(2, 3, 4)$ and perpendicular to the plane given by $3x + 2y - z = 6$ are

$$x = 2 + at, \quad y = 3 + bt, \quad z = 4 - t,$$

then $a + b =$

(a) 1

(b) 5

(c) -3

(d) 3

(e) -5

15. Consider the function $f(x, y) = \left(x^2 + y^2\right)^{2/3}$. Then

(a) $f_x(0, 0) = 0$.

(b) $f_x(1, 1) \neq f_x(1, -1)$.

(c) $f_x(0, 0)$ and $f_y(0, 0)$ do not exist.

(d) $f_x(2, 2) = \dfrac{3}{4}$.

(e) $f_y(1, 1) = \dfrac{4}{3}$.

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	Inside $u \in (0, 25]$, so $\ln u \in (-\infty, \ln 25] = (-\infty, 2\ln 5]$.
2	**(b)**	$-F_x/F_z = -\frac{6xz-2xy^2}{3x^2+6z^2+3y} = \frac{2x(y^2-3z)}{3(x^2+2z^2+y)}$.
3	**(a)**	Completing squares: $(x-2)^2 - 2(y-1)^2 + (z-3)^2 = k+11$; $k=-11$ gives an elliptic cone.
4	**(d)**	$\mathbf{n} = \langle 6, 4, -1\rangle$, so $\cos\theta = \frac{\lvert-1\rvert}{\sqrt{53}} = \frac{1}{\sqrt{53}}$.
5	**(d)**	$\nabla F(0,2,2) = \langle 8, 1, -1\rangle$, so $b=1$, $c=-1$ and $b+c=0$.
6	**(d)**	The domain contains no open disk around $(0,0)$, so by the textbook definition the limit does not exist.
7	**(e)**	$\nabla f = \langle 0, -2\rangle$, $\mathbf{u} = \langle \frac{3}{5}, -\frac{4}{5}\rangle$, so $D_{\mathbf{u}} f = \frac{8}{5}$.
8	**(e)**	$R = 6$, $R_i' = R^2/R_i^2$: $dR = 0.36(0.5) + 0.16(-2) = -0.14$.
9	**(d)**	Each of $f_{xzz}, f_{zxz}, f_{zzx}$ equals $-\frac{1}{z^2} = -1$ at $(1,1,1)$; sum $= -3$.
10	**(d)**	$\frac{\lvert 4+8+4-5\rvert}{\sqrt{4+1+1}} = \frac{11}{\sqrt{6}}$.
11	**(d)**	Plane $-4(x-1) + 4(y+1) + 8(z-4) = 0 \Rightarrow x - y - 2z + 6 = 0$; $a+b+c = 3$.
12	**(a)**	$w_s = (y+z)\cos t + (x+z)\sin t$; at $s=1, t=2\pi$ this is 2π.
13	**(e)**	$\nabla f(0,3) = \langle -\sqrt{3}, 6\rangle$, so the maximum is $\lVert\nabla f\rVert = \sqrt{39}$.
14	**(b)**	Direction $= \mathbf{n} = \langle 3, 2, -1\rangle$, so $a=3$, $b=2$ and $a+b=5$.
15	**(a)**	$f_x(0,0) = \lim \frac{\lvert h\rvert^{4/3}}{h} = 0$; the other statements fail by direct computation.